ModelIt!

Teacher's Guide

The Invisible War: How Your Immune System Fights Back

Your Body's Army Against Invaders

Quick Reference

Grade Level	Grade 07L2
NGSS Standard	MS-LS1-3, MS-LS1-8
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-3, MS-LS1-8

Use argument supported by evidence for how the body is a system of interacting subsystems composed of groups of cells. Gather and synthesize information that sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students use model evidence to argue how the body's immune subsystems interact to fight pathogens.
Disciplinary Core Idea	**LS1.A Structure and Function** — The body is a system of interacting subsystems composed of groups of cells that work together to fight infection and maintain health.
Crosscutting Concept	**Systems and System Models** — Students model the immune system as interacting subsystems where white blood cells, antibodies, and memory cells work together.

CAST Testing Alignment

MULTIPLE CHOICE:

A vaccinated person and an unvaccinated person are both exposed to the same virus. The vaccinated person shows mild symptoms while the unvaccinated person becomes severely ill. Which best explains this difference?

CONSTRUCTED RESPONSE:

Using your immune system model, explain how white blood cells and antibodies work together as interacting subsystems to fight a pathogen. Include how vaccination changes this interaction.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Your body is constantly under attack from pathogens, bacteria, viruses, and other microorganisms that can cause disease. The immune system is a complex network of cells, tissues, and organs that work together to defend the body, acting as interacting subsystems within the larger body system.

First Line of Defense

The body's first defenses are physical barriers: skin blocks most pathogens, mucus traps them in the nose and throat, and stomach acid destroys many that are swallowed. These are non-specific defenses that work against all types of invaders.

White Blood Cell Response

When pathogens breach physical barriers, white blood cells are the first immune cells to respond. Neutrophils arrive within hours, engulfing and destroying invaders. Macrophages follow, cleaning up debris and presenting pathogen fragments to other immune cells to trigger a targeted response.

Antibody Production

B-cells produce antibodies: Y-shaped proteins that lock onto specific pathogen molecules called antigens. Each antibody is custom-made for one pathogen type. The first time the body encounters a pathogen, antibody production takes 7-10 days. This delay is why new infections make you sick.

Immune Memory and Vaccines

After fighting an infection, the immune system creates memory cells that remember the pathogen's identity. If the same pathogen returns, antibody production begins within hours instead of days. Vaccines exploit this by introducing harmless versions of pathogens to trigger memory cell formation without causing disease.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify pathogen load, vaccine status, white blood cell response, antibody production, and infection severity as key components of the immune system.

E **Establish Relationships**
Students determine that pathogen load drives infection severity positively while vaccine status and immune responses reduce it through negative relationships.

V **Visualize & Model**
Students build a model showing how the immune system's interacting subsystems respond to pathogen invasion with and without prior vaccination.

E **Evaluate Outcomes**
Students run vaccinated, unvaccinated, and low-exposure scenarios to compare immune response timelines and infection outcomes.

R **Revise & Extend**
Students add memory cell formation or fever response to explore how the immune system adapts and strengthens over time.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, inside your body, there's a war happening. And you're winning."*

Watch For

Show a dramatic visualization of immune cells attacking pathogens.

Slide 2: Learning Objectives

3 min

Say: *"Today we're building a model of the most sophisticated defense system on Earth: YOUR immune system."*

Watch For

Have students read objectives. Pre-teach 'pathogen' and 'antibody.'

Slide 3: Big Question

3 min

Say: *"You've been sick before. But how does your body know something doesn't belong? And how does it fight back?"*

Watch For

Quick poll: Who has been sick in the last year? What happened? How did you recover?

Slide 4: LEVER Framework

2 min

Say: *"We'll model the immune system as a set of interacting parts that work together."*

Watch For

Preview each LEVER step. Emphasize the idea of SUBSYSTEMS working together.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key players in this invisible war? What can we control vs. what responds automatically?"*

Watch For

Guide sorting of external vs. internal components. Discuss why vaccine status is an input.

Slide 6: Activity 2: Connections

8 min

Say: *"When pathogen load goes up, what happens to white blood cell response? And how does that affect how sick you get?"*

Watch For

Help students identify positive and negative relationships. Trace the cascade.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's infect a vaccinated person and an unvaccinated person with the same virus. What happens?"*

Watch For

Students predict first, then run simulations. Compare the graphs side by side.

Slide 8: Discoveries

5 min

Say: *"So why do vaccinated people get less sick? What did our model reveal about the TIMING of antibody production?"*

Watch For

Lead discussion about immune memory. Connect to real-world vaccination data.

Slide 9: STEM Challenge

12 min

Say: *"You now understand the science. Can you explain it to someone who doesn't?"*

Watch For

Groups design public health campaigns using model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Pathogen Load, Vaccine Status
- **INTERNAL:** White Blood Cell Response, Antibody Production, Infection Severity

Relationships

- **Pathogen Load to Infection Severity:** POSITIVE (+) — More pathogens entering the body means the infection can spread faster and cause more damage before the immune system catches up.
- **Vaccine Status to Antibody Production:** POSITIVE (+) — When the immune system has been vaccinated, it already has memory cells ready to produce antibodies immediately upon infection.
- **White Blood Cell Response to Infection Severity:** NEGATIVE (-) — A stronger white blood cell response destroys more pathogens early, reducing how severe the infection becomes.
- **Antibody Production to Infection Severity:** NEGATIVE (-) — Antibodies specifically target and neutralize pathogens, directly reducing infection severity.
- **Pathogen Load to White Blood Cell Response:** POSITIVE (+) — More pathogens trigger a larger immune response as the body detects the growing threat.

Key Discoveries

- The immune system has two lines of defense: fast general response (white blood cells) and slow targeted response (antibodies)
- Vaccines work by giving the immune system a head start, so antibody production begins immediately upon infection
- Higher pathogen loads overwhelm the immune system faster, making the white blood cell response critical in the first hours
- The immune system remembers past infections, which is why you rarely get the same illness twice

Multiple Choice Post-Assessment Answer Key

1. A vaccinated person and an unvaccinated person are both exposed to the same high dose of a virus. The vaccinated person recovers in two days with mild symptoms, while the unvaccinated person is sick for ten days. Based on the immune system model, which explanation best accounts for this difference?

Correct: B) The vaccinated person's memory cells enable rapid antibody production, neutralizing the pathogen before it can spread widely

Correct! Vaccination creates memory cells that recognize the pathogen immediately upon re-exposure. This allows antibody production to begin within hours instead of the 7-10 days it takes for a first encounter, reducing infection severity dramatically.

2. In the immune system model, when Pathogen Load is set to high and Vaccine Status is off, Infection Severity rises rapidly before White Blood Cell Response can contain the invasion. Which relationship in the model best explains this outcome?

Correct: C) Pathogen Load has a positive relationship with Infection Severity, and antibody production is delayed without prior vaccination

Correct! Pathogen Load positively drives Infection Severity, meaning more pathogens cause worse illness. Without vaccination, there are no memory cells, so the targeted antibody response takes 7-10 days to ramp up, allowing the infection to spread during that critical delay.

3. A student claims that white blood cells and antibodies perform the same function in the immune system. Using the model, which response best corrects this misconception?

Correct: A) White blood cells provide a fast, general response against any invader, while antibodies are custom-built proteins that specifically target one pathogen type

Correct! The immune system operates as interacting subsystems. White blood cells provide the immediate, non-specific first response, attacking any invader within hours. Antibodies are produced by B-cells and are custom-designed to target specific pathogens, but take days to produce during a first infection.

4. In the model, setting Pathogen Load to low with Vaccine Status off still results in low Infection Severity. Which explanation best accounts for why the body can handle a small invasion without prior vaccination?

Correct: B) The non-specific white blood cell response is sufficient to contain a small number of pathogens before they can multiply significantly

Correct! When pathogen load is low, the fast-acting white blood cell response can engulf and destroy the invaders before they multiply to overwhelming numbers. The body does not need the slower antibody response because the initial defense handles the small threat.

5. A community has 90% vaccination rates for a particular virus. A student who cannot be vaccinated due to a medical condition is protected from the disease. Which concept from the model best explains this protection?

Correct: B) When most people in a community are immune, the pathogen cannot spread easily enough to reach vulnerable individuals, which is known as herd immunity

Correct! Herd immunity occurs when enough people in a population are immune that the pathogen cannot find enough susceptible hosts to sustain transmission. This protects individuals who cannot be vaccinated because the pathogen simply cannot reach them through the immune community.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Vaccines give you the disease"	Vaccines contain weakened, inactivated, or partial versions of pathogens that CANNOT cause the full disease. Some people feel mild side effects (sore arm, low fever) because the immune system is RESPONDING to the vaccine and building defenses, not because they're sick.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Antibiotics kill viruses"	Antibiotics only work against bacteria by disrupting their cell walls or reproduction. Viruses are not cells, so antibiotics have no effect on them. That's why doctors don't prescribe antibiotics for the flu or common cold.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If you're healthy, you don't need vaccines"	Even healthy people can get severely ill from certain diseases. Vaccines also protect vulnerable people around you (babies, elderly, immunocompromised) through herd immunity. Your vaccination protects your community, not just you.